

3D chromatin differential accessibility and arrangement during human-chimpanzee evolution

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Introduction

Variation in gene regulation has emerged as an important mechanism for producing the broad diversity of life present on Earth today (King and Wilson, 1975; Levine and Tjian, 2003; Carroll, 2005; Wray, 2007). How these gene regulatory differences arise is a fundamental question in evolutionary biology. Changes to tissue-specific regulatory sequences are thought to be particularly important drivers of evolutionary change because they can produce targeted gene expression differences that are unlikely to have broad detrimental effects (Wray, 2007; Carroll, 2008). Indeed, many studies have demonstrated important phenotypic changes resulting from the creation, deletion, or modification of regulatory sequences including tissue-specific enhancers in diverse organisms (Olds and Sibley, 2003; Gompel et al., 2005; Chan et al., 2010; McLean et al., 2011; Wittkopp and Kalay, 2012).

However, does evolution ever act through changes that produce broader gene regulatory differences? One intriguing mechanism to produce such changes is to modify larger three-dimensional chromatin organizational structures. Eukaryotic genomes are organized into topologically associating domains (TADs), units of chromatin within which there is high contact frequency but between which contact occurs relatively infrequently (Dixon et al., 2012; Symmons et al., 2014; Zhan et al., 2017). Boundaries between TADs act as insulators and are established by proteins including CCCTC-binding factor (CTCF), cohesin, and RNA polymerase II (Dixon et al., 2012; Phillips-Cremins et al., 2013; Rao et al., 2014; Van Bortle et al., 2014). TADs and their boundaries therefore help establish separate genomic regions within which regulatory regions can

influence certain genes but are insulated from regulating other nearby genes. Disrupting TAD structure—perhaps via addition, deletion, or modification of a TAD boundary—therefore has the potential to bring multiple new regulatory regions in contact with multiple new genes and result in substantial gene expression and downstream phenotypic effects.

Indeed, there is considerable evidence that modifying TAD structure can cause gene misexpression and disease in humans (Giorgio et al., 2015; Lupiáñez et al., 2015; Franke et al., 2016; Hnisz et al., 2016; Lupiáñez et al., 2016). For example, experiments in mice that recreate TAD boundary inversions found in patients with limb malformations result in the misexpression of nearby important limb development genes, leading to similar limb malformations in the mice (Lupiáñez et al., 2015). Others have shown that a subset of adult-onset leukodystrophy patients have a deletion that removes a TAD boundary and enhancer, resulting in ectopic regulation of *LMNB1* by another nearby enhancer (Giorgio et al., 2015).

However, little work has investigated chromatin rearrangements between species as a potential mechanism for beneficial evolutionary change. Indeed, most comparative chromatin organization studies have focused on the high conservation of TADs and their boundaries across species (Dixon et al., 2012; Dekker and Heard, 2015; Vietri Rudan et al., 2015; Krefting et al., 2018; Fudenberg and Pollard, 2019). Still, studies such as the investigation of a large inversion that causes TAD rearrangement and contributes to the adaptive masculinization of ovotestes in female moles demonstrate that chromatin rearrangements can confer evolutionary benefits (Real et al., 2020). Furthermore, there has been limited study of chromatin organization in non-human primates (Vietri Rudan et al., 2015; Krefting et al., 2018; Lazar et al., 2018), and few studies to-date have specifically compared genome-wide chromatin organization between humans and our closest extant relatives, chimpanzees (Eres et al., 2019; Fudenberg and Pollard, 2019; Yang et al., 2019).

Here, we explore the role of three-dimensional chromatin rearrangements in human evolution using our panel of human or chimp diploid, human or chimpanzee autotetraploid (same-species), and human-chimpanzee allotetraploid (different-species) induced pluripotent stem cell (iPSC) lines described in Song, Grant, Behrens *et al.* 2021. Specifically, we conduct ATAC-sequencing and Hi-C on our iPSCs to investigate genome-wide species differences in chromatin

accessibility and three-dimensional organization, and their impact on nearby gene expression. We identify a small set of dramatic chromatin rearrangements between human and chimpanzee iPSCs which are prime candidates for further investigation into their evolutionary-significance. Finally, we use CRISPR-Cas9 editing to experimentally test one of these candidates, which contains several human-specific features and is near an evolutionarily-intriguing gene.

Results

ATAC peaks differentially-accessible between human and chimpanzee chromosomes are enriched for CTCF binding motifs

To investigate chromatin accessibility differences between humans and chimpanzees, we carried out ATAC-sequencing on the human-chimpanzee allotetraploid iPS cell line H1C1 (generated and described by Song, Grant, Behrens *et al.* 2021). The resulting 210,461 ATAC peaks are evenly distributed between the two species, with 107,097 on human chromosomes and 103,364 on chimpanzee chromosomes at a false discovery rate (FDR) of 5%. Of these, 29,683 peaks are differentially accessible between the two species, with 12,718 more accessible on human chromosomes and 16,964 more accessible on chimpanzee chromosomes (5% FDR).

We next tested whether certain transcription factor (TF) binding motifs are more common in regions that are differentially-accessible between human and chimpanzee chromosomes. We found that motifs for CTCF and its paralog Brother Of the Regulator of Imprinted Sites (BORIS), which is encoded by the gene *CTCF*, are the only TF binding motifs that are statistically significantly enriched in ATAC peaks that are differentially-accessible between human and chimpanzee chromosomes ($p = 0.0459$ for CTCF and $p = 0.0017$ for BORIS). We carried out the same TF binding motif enrichment analysis on progressively smaller sets of ATAC peaks with progressively more stringent statistical significance values as a proxy for peaks with increasingly more divergent accessibility between the human and chimpanzee chromosomes. We found that the more divergent the accessibility of the peak, the greater the enrichment of CTCF and BORIS (Table 5.1). Specifically, after computing enrichment ratios by dividing the number of ATAC peaks with a binding motif by the number of orthologous background sequences from the opposite species with a binding motif, we see an enrichment of 1.18 compared to 1.57 for the sets of ATAC

peaks with significance <0.05 and $<1 \times 10^{-8}$, respectively. For BORIS, the corresponding enrichment ratios are 1.18 and 1.46, respectively.

ATAC peak significance level	# ATAC peaks in the tested set	% target sequences with motif (N)	% background sequences with motif (N)	TF motif enrichment corrected p-value	TF motif enrichment ratio
CTCF					
<0.05	6,427	14.7% (947)	12.46% (801)	0.0459	1.18
$<1 \times 10^{-3}$	3,452	13.5% (466)	10.14% (350)	0.0041	1.27
$<1 \times 10^{-5}$	2,031	14.4% (292)	10.20% (207)	0.0131	1.41
$<1 \times 10^{-8}$	1,120	14.6% (164)	9.29% (104)	0.0262	1.57
BORIS					
<0.05	6,427	19.1% (1227)	15.96% (1026)	0.0017	1.12
$<1 \times 10^{-3}$	3,452	18.2% (629)	13.88% (479)	0.0005	1.31
$<1 \times 10^{-5}$	2,031	19.2% (390)	13.98% (284)	0.0044	1.37
$<1 \times 10^{-8}$	1,120	20.3% (227)	13.80% (155)	0.0243	1.46

Table 5.1: CTCF and BORIS binding motifs are enriched in species-differential ATAC-peaks

Transcription factor binding motif enrichment analysis results in species-differentially accessible ATAC peaks between the human and chimpanzee chromosomes in allotetraploid iPSCs. Only species-differentially accessible ATAC peaks predicted to regulate genes differentially expressed at least in part due to *cis*-regulatory differences were included in the analysis (Song et al., 2021). Orthologous sequences from the other, non-enriched, species were used as background. Increasingly stringent p-value thresholds for differential ATAC peak significance were tested as a proxy for increasingly divergent chromatin accessibility between the human and chimpanzee chromosomes. The TF binding motif enrichment ratio increases with increasingly more significant ATAC peak thresholds and was computed by dividing the number of target sequences by the number of background sequences. The reported TF binding motif enrichment p-value was calculated and adjusted using Bonferroni correction by HOMER (Heinz et al., 2010) (see Methods).

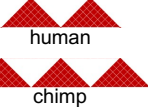
Although we report the enrichment of binding motifs both for CTCF and its paralog BORIS, we consider CTCF to be the biologically relevant factor in this case. This is because our ATAC-sequencing data was generated from iPS cells, but BORIS is highly expressed only in testicular germ cells and cancer cells (Loukinov et al., 2002; Dougherty et al., 2008; Renaud et al., 2011; Liu et al., 2017). In contrast, CTCF is a ubiquitously expressed, highly-conserved core architectural protein important for establishing three-dimensional genome organization in eukaryotes and is consistently enriched at the boundaries between topologically-associated domains (Dixon et al., 2012; Phillips-Cremins et al., 2013; Rao et al., 2014; Van Bortle et al.,

2014). Note that since CTCF is an insulator protein that largely affects the arrangement of nearby chromatin regions, the final transcription factor binding analysis detailed above was carried out using only those ATAC peaks likely to influence genes that are differentially expressed between human and chimpanzee iPSCs due at least in part to *cis*-regulatory changes according to the gene classifications in Song, Grant, Behrens *et al.* 2021.

A small fraction of TADs are differential between humans and chimpanzees and can be classified based on chromatin rearrangement severity

We hypothesized that the pattern of CTCF binding motif enrichment in species-differentially-accessible ATAC peaks may suggest there are evolutionarily-interesting three-dimensional chromatin arrangement differences between humans and chimpanzees. To test this hypothesis, we carried out Hi-C on two human diploid (H1 and H2), two chimpanzee diploid (C1 and C2), two human autotetraploid (H1H1a and H1H1b), two chimpanzee autotetraploid (C1C1b and C1C1e), and four human-chimpanzee allotetraploid (H1C1h, H1C1k, H2C2a, and H2C2d) cell lines. We carried out most analyses in parallel by comparing (1) single-species human diploids and autotetraploids versus chimpanzee diploids and autotetraploids and (2) the human chromosome versus the chimpanzee chromosome within allotetraploids. We identified 8,259 conserved and 871 differential TADs in single-species human and chimpanzee iPSCs, and similarly identified 7,818 conserved and 621 allele-specific TADs within allotetraploids.

We next used TADcompare (Cresswell and Dozmorov, 2020) to classify the differential and allele-specific TADs into five categories: (1) **Split**: Two human TADs are a single TAD in chimpanzee, (2) **Merge**: The opposite of split; two chimpanzee TADs are a single TAD in human, (3) **Complex**: Human and chimpanzee TADs have entirely different structures, (4) **Shifted**: A TAD boundary in one species is shifted by 1-5 bins in the other species, and (5) **Strength Change**: A TAD boundary is present in both species but has a differential boundary score with magnitude greater than 2 (Table 5.2). For both single-species and allotetraploid comparisons, Strength Change is the most common non-differential category, composing 8.31% (N = 759) and 6.30% (N = 532) of all non-differential TADs, respectively. The remaining four categories contain comparatively few TADs, ranging from 16 in the allotetraploid Split category to 48 in the single-species Shifted category.

	Split	Merge	Complex	Shifted	Strength change	Non-differential
						
Single-species diploid and autotetraploid iPSCs						
% of total TADs in category (N)	0.21% (19)	0.26% (24)	0.23% (21)	0.53% (48)	8.31% (759)	90.46% (8,259)
% with <i>cis</i> gene DE between species (N)	79% (15)	75% (18)	76% (16)	71% (34)	68% (518)	52% (4,281)
Human-chimpanzee allotetraploid iPSCs						
% of total TADs in category (N TADs)	0.18% (16)	0.27% (23)	0.21% (18)	0.38% (32)	6.30% (532)	92.64% (7,818)
% with <i>cis</i> gene DE between species (N)	81% (13)	78% (18)	83% (15)	72% (23)	72% (383)	47% (3,691)

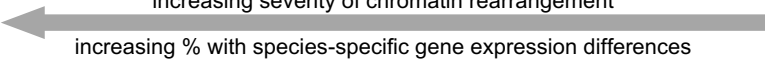


Table 5.2: More severe chromatin rearrangements between human and chimpanzee iPSCs are more likely to contain differentially expressed genes

Categorization of species-differential TADs in single-species iPSCs (top) and allele-specific TADs in allotetraploid iPSCs (bottom) based on Hi-C. The Split, Merge, and Complex categories are more severe rearrangements compared to the Shifted and Strength Change categories, as shown by the schematic below the category name. A higher percentage of TADs within the more severe rearrangement categories contain genes differentially expressed due to *cis*-regulatory differences between human and chimpanzee iPSCs (Song et al., 2021), as indicated by the gray arrow.

Genomic regions with more severe chromatin structural differences between human and chimpanzee iPSCs contain more differentially expressed genes

To investigate the biological significance of these differential TAD categories, we determined the percentage of TADs in each category that contain at least one gene that is differentially expressed between humans and chimpanzees due at least in part to *cis*-regulatory differences according to the gene classifications in Song, Grant, Behrens *et al.* 2021. This revealed that compared to the two categories representing minor changes (Shifted and Strength Change), TADs in the three categories representing major chromatin rearrangements (Split, Merge, and Complex) are more likely to contain a gene that is differentially expressed due to *cis*-regulatory

differences (Table 5.2). Indeed, for diploids and autotetraploids, only 68% and 71% in the Strength Change and Shifted categories compared to 79%, 75%, and 76% in the Split, Merge, and Complex categories, respectively, contain at least one gene that is differentially expressed between humans and chimpanzees due to *cis*-regulatory differences. The allotetraploids follow a similar pattern, with 72% and 72% compared to 81%, 78%, and 83%, respectively. Since the number of differentially expressed genes within a given region is positively correlated with its chromatin rearrangement severity, we conclude that the TAD category classifications are capturing biologically-meaningful species differences. Furthermore, the majority of TADs in the Split, Merge, and Complex categories are identical between our single-species and allotetraploid comparisons, giving further confidence to our categorization and highlighting that tetraploidization does not substantially affect TAD structure.

We also tested several other hypotheses that were not supported. First, differential TAD boundaries are not enriched for human conserved deletions (hCONDELs) (McLean et al., 2011), human accelerated regions (HARs) (Pollard et al., 2006; Hubisz and Pollard, 2014; Girsakis et al., 2021), ClinVar variants (Landrum et al., 2018), or any transcription factor binding motifs including CTCF. Second, considering only TADs with species-differential border strength (i.e. TADs in the Strength Change category), TAD border strength magnitude difference was not correlated with gene expression magnitude difference of differentially-expressed genes within those TADs.

Prioritizing candidate regions with chromatin structural differences between human and chimpanzee iPSCs

Genomic regions in the Split, Merge, and Complex categories represent those with the most dramatic chromatin rearrangements between humans and chimpanzees and are therefore intriguing candidates for further experimentation. Given the CTCF binding motif enrichment in our species-differential ATAC peaks and the importance of CTCF for establishing TADs, we prioritized regions with species-specific CTCF motifs located at species-specific TAD borders. We identified only one such region, and it is on chromosome 20 in the Complex category. It contains a human-specific CTCF motif at a human-specific TAD border that creates a human-specific sub-TAD within a larger TAD conserved between humans and chimpanzees (Fig. 5.1A). There is a single G→A substitution at a conserved nucleotide in the primate lineage that creates a new CTCF

binding motif in humans at chr20:61,120,827-61,120,846 (hg38) (Fig. 5.1B). This region is particularly promising because the human-specific TAD and CTCF binding motif also overlap a human-specific ATAC peak in our data as well as a CTCF ChIP-sequencing peak from external data in human embryonic stem cells (Zhang et al., 2020), supporting it as a functional binding motif. Furthermore, the human-specific TAD boundary and CTCF binding motif is ~131 kb upstream of the 5' end of the *Cadherin 4* (*CDH4*) gene (Fig. 5.1A). *CDH4* shows 3.14 fold higher expression—with 1.93 fold due to *cis*-regulatory differences and the remaining due to *trans*-regulatory differences—in our human compared to chimpanzee iPSCs according to RNA-sequencing data from Song, Grant, Behrens *et al.* 2021.

Ablating the human-specific CTCF binding motif at a human-specific TAD border on chromosome 20 near *CDH4* using CRISPR-Cas9

We next tested whether this human-specific CTCF binding motif on chromosome 20—and potentially the human-specific sub-TAD it may influence—is responsible for increased *CDH4* expression in human iPSCs. To do this, we used CRISPR-Cas9 to ablate the human-specific CTCF binding motif in the human diploid iPS cell line H2 (Fig. 5.1B). For comparison, we used the same CRISPR gRNA to target the orthologous region in the chimpanzee diploid iPS cell line C2. For unedited controls, we subjected the same human and chimpanzee diploid cell lines to identical editing conditions using a scrambled version of the CRISPR gRNA.

Across three replicates, we achieved an average editing efficiency of 89.13% (SD = 0.36%) in human iPSCs and 83.57% (SD = 1.86%) in chimpanzee iPSCs as assessed by amplicon sequencing. Importantly, however, high editing efficiency does not guarantee that the Cas9-induced lesions actually ablated the human-specific CTCF binding motif in human iPSCs. We therefore confirmed there was successful ablation of the CTCF binding motif in 76.73% (SD = 0.75%) of all human iPSCs and aberrant creation of a new CTCF binding motif in only 0.22% (SD = 0.02%) of all chimpanzee iPSCs.

The edit types were as expected from non-homologous end-joining (NHEJ) repair of Cas9-induced double-strand breaks, with insertions or deletions composing an average of 80.86% (SD = 1.16%) and 74.66% (SD = 2.35%) of all successful edits across replicates in human and

223 chimpanzee iPSCs, respectively (Table 5.3). The majority of the remaining successful edits (mean
 224 = 15.67%, SD = 0.76% in human iPSCs; mean = 22.12%, SD = 2.64% in chimpanzee iPSCs) had
 225 from 1 to 5 base pair changes, with edits of 6 or more base pair changes composing the remaining
 226 ~3-4%.

	Edit type	Percent with edit	Indel length
Human edited iPSCs			
CCCCCTGCTCGCACTTCTCTCTCCTGCCA	WT (unedited)	11.23%	0
CCCCCTGCTCGCACTTCTCTCTCCTGCCA	Deletion	5.54%	-15
CCCCCTGCTCGCACTTCTCTCTCCTGCCA	Insertion	4.89%	1
CCCCCTGCTCGCACTTCTCTCTCCTGCCA	Deletion	2.43%	-8
CCCCCTGCTCGCACTTCTCTCTCCTGCCA	Deletion	1.90%	-1
CCCCCTGCTCGCACTTCTCTCTCCTGCCA	Deletion	1.20%	-22
CCCCCTGCTCGCACTTCTCTCTCCTGCCA	Deletion	0.80%	-23
CCCCCTGCTCGCACTTCTCTCTCCTGCCA	Deletion	0.75%	-18
CCCCCTGCTCGCACTTCTCTCTCCTGCCA	Deletion	0.75%	-10
CCCCCTGCTCGCACTTCTCTCTCCTGCCA	Deletion	0.50%	-25
CCCCCTGCTCGCACTTCTCTCTCCTGCCA	Deletion	0.43%	-25
CCCCCTGCTCGCACTTCTCTCTCCTGCCA	3 base changes	0.40%	0
CCCCCTGCTCGCACTTCTCTCTCCTGCCA	Deletion	0.35%	-16
CCCCCTGCTCGCACTTCTCTCTCCTGCCA	Deletion	0.34%	-17
CCCCCTGCTCGCACTTCTCTCTCCTGCCA	Deletion	0.33%	-12
CCCCCTGCTCGCACTTCTCTCTCCTGCCA	Deletion	0.32%	-31
CCCCCTGCTCGCACTTCTCTCTCCTGCCA	Deletion	0.31%	-3
CCCCCTGCTCGCACTTCTCTCTCCTGCCA	Deletion	0.31%	-27
CCCCCTGCTCGCACTTCTCTCTCCTGCCA	Deletion	0.30%	-19
CCCCCTGCTCGCACTTCTCTCTCCTGCCA	Insertion	0.30%	5
CCCCCTGCTCGCACTTCTCTCTCCTGCCA	Deletion	0.29%	-11
CCCCCTGCTCGCACTTCTCTCTCCTGCCA	Deletion	0.27%	-2
CCCCCTGCTCGCACTTCTCTCTCCTGCCA	Deletion	0.24%	-23
CCCCCTGCTCGCACTTCTCTCTCCTGCCA	Deletion	0.23%	-13
CCCCCTGCTCGCACTTCTCTCTCCTGCCA	2 base changes	0.21%	0
CCCCCTGCTCGCA	Deletion	0.21%	-32
CCCCCTGCTCGCACTTCTCTCTCCTGCCA	Deletion	0.20%	-24
CCCCCTGCTCGCACTTCTCTCTCCTGCCA	Deletion	0.20%	-16
Chimpanzee edited iPSCs			
CCCCCTGCTCGCACTTCTCTCTCCTGCCG	WT (unedited)	14.30%	0
CCCCCTGCTCGCACTTCTCTCTCCTGCCG	Insertion	7.38%	1
CCCCCTGCTCGCACTTCTCTCTCCTGCCG	Deletion	3.70%	-15
CCCCCTGCTCGCACTTCTCTCTCCTGCCG	Deletion	3.03%	-1
CCCCCTGCTCGCACTTCTCTCTCCTGCCG	Deletion	0.90%	-8
CCCCCTGCTCGCACTTCTCTCTCCTGCCG	Deletion	0.86%	-22
CCCCCTGCTCGCACTTCTCTCTCCTGCCG	Deletion	0.60%	-5
CCCCCTGCTCGCACTTCTCTCTCCTGCCG	Deletion	0.56%	-2
CCCCCTGCTCGCACTTCTCTCTCCTGCCG	Deletion	0.44%	-3
CCCCCTGCTCGCACTTCTCTCTCCTGCCG	Deletion	0.39%	-18
CCCCCTGCTCGCACTTCTCTCTCCTGCCG	Deletion	0.33%	-23
CCCCCTGCTCGCACTTCTCTCTCCTGCCG	Deletion	0.29%	-24
CCCCCTGCTCGCACTTCTCTCTCCTGCCG	3 base changes	0.29%	0
CCCCCTGCTCGCACTTCTCTCTCCTGCCG	Deletion	0.27%	-25
CCCCCTGCTCGCACTTCTCTCTCCTGCCG	Deletion	0.26%	-11
CCCCCTGCTCGCACTTCTCTCTCCTGCCG	Deletion	0.25%	-14
CCCCCTGCTCGCACTTCTCTCTCCTGCCG	Deletion	0.24%	-10
CCCCCTGCTCGCACTTCTCTCTCCTGCCG	Deletion	0.23%	-25
CCCCCTGCTCGCACTTCTCTCTCCTGCCG	Deletion	0.21%	-12
CCCCCTGCTCGCACTTCTCTCTCCTGCCG	2 base changes	0.21%	0
CCCCCTGCTCGCACTTCTCTCTCCTGCCG	Deletion	0.21%	-2
CCCCCTGCTCGCACTTCTCTCTCCTGCCG	Deletion	0.20%	-20

227 **Table 5.3: Successful ablation of the human-specific CTCF binding motif using CRISPR-Cas9**
 228 Edits occurring at a frequency $\geq 0.2\%$ based on amplicon sequencing in a representative replicate from human edited
 229 iPSCs and chimpanzee edited iPSCs. The human-specific CTCF binding motif is highlighted in light purple, and the
 230 SNP within it is highlighted in darker purple.

***CDH4* gene expression after ablating the human-specific CTCF binding motif on chromosome 20**

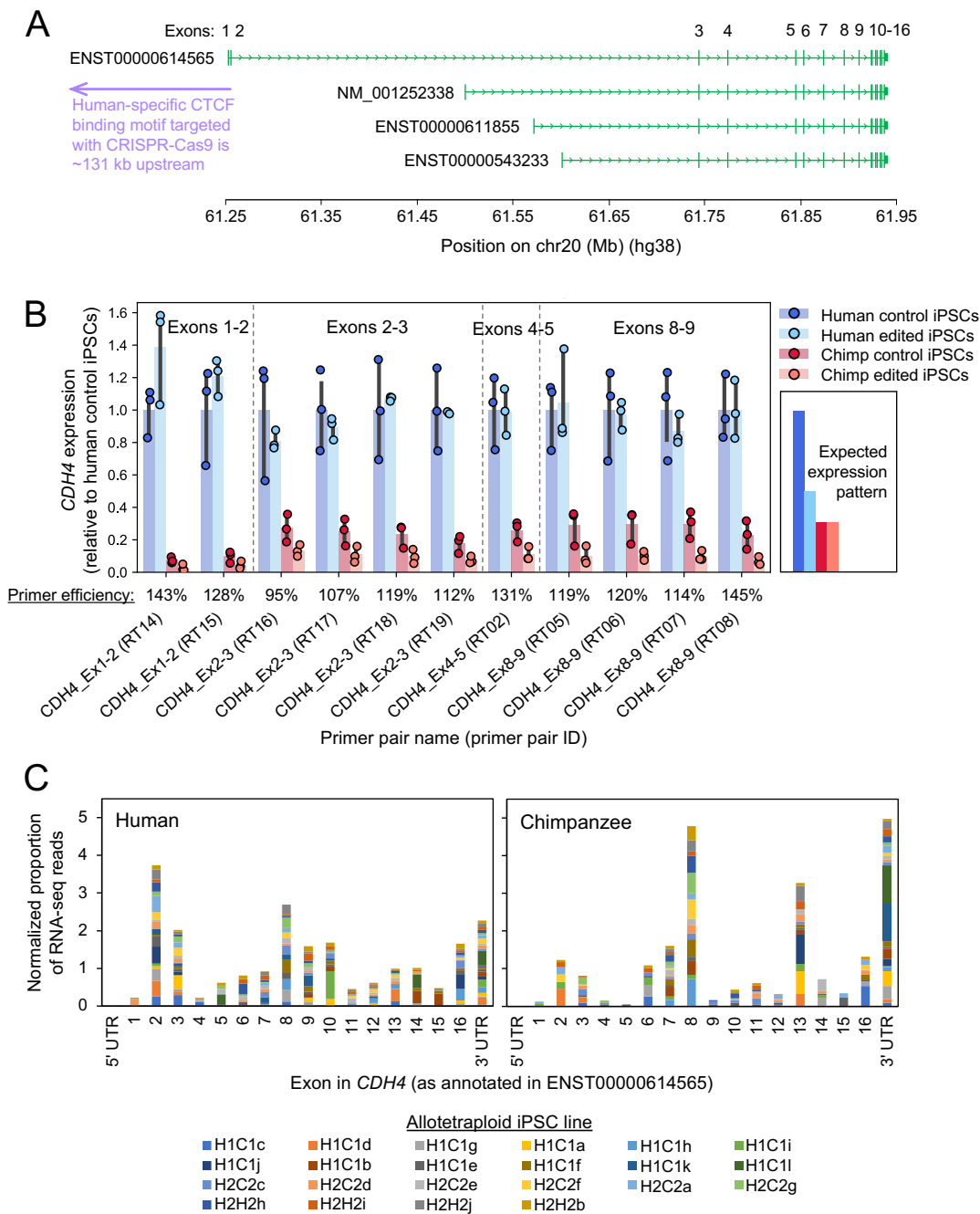
We predicted that ablating the human-specific CTCF binding motif in human cells may create a more chimpanzee-like three-dimensional genome organization and perhaps therefore lower the expression of nearby gene *CDH4* to more chimpanzee-like levels (Fig. 5.2B). Conversely, we expected *CDH4* expression to remain constant after editing the orthologous region in chimpanzees since the region is not a CTCF binding site or TAD border in chimpanzees (Fig. 5.2B).

To test our hypothesis, we used RT-qPCR to assess *CDH4* expression in our edited and unedited human and chimpanzee iPSCs. Since there are multiple *CDH4* isoforms annotated in humans (Fig. 5.2A), we designed primers spanning four different *CDH4* exon junctions. It was not possible to achieve the ideal primer efficiency of 90-110% for all primer pairs and we therefore include results from multiple primer pairs for most exon junctions. As expected from our RNA-sequencing data (Song et al., 2021), we consistently found ~3 fold higher expression in unedited human iPSCs compared to unedited chimpanzee iPSCs for all primer pairs ($p < 0.05$ by Student's t-test for all primer pairs) (Fig. 5.2B). However, we also find two unexpected but consistent patterns: (1) *CDH4* expression does not statistically significantly change in human cells after ablating the human-specific CTCF site, and (2) *CDH4* expression decreases by ~2-3 fold after editing the orthologous region in chimpanzee cells ($p < 0.05$ by Student's t-test for all primer pairs except the two spanning exons 1-2) (Fig. 5.2B). Please see the Discussion section for potential explanations.

Chromosomal abnormalities after CRISPR-Cas9

Additionally, since this region is located on chromosome 20q, we assessed chromosome 20 dosage relative to other chromosomes using qPCR for all replicates of all conditions included in our CRISPR-Cas9 experiment (except for one unedited chimpanzee replicate which had poor DNA quality after extraction). This was particularly important since chromosome 20q aberrations are commonly acquired and found in ~20% of human pluripotent stem cells worldwide (Amps et al., 2011). Of particular concern was the common isochromosome 20q (iso20q) aberration which

259 results in one normal chromosome 20 (one p and one q arm) and one abnormal chromosome 20
260 with two q arms (Vitillo et al., 2023).



261 **Figure 5.2: *CDH4* gene expression after CRISPR-Cas9 editing and *CDH4* exon usage in allotetraploid iPSCs**

262 **A.** *CDH4* transcripts annotated in the human genome. Exons are labeled (top) according to transcript
263 ENST00000614565. Image is adapted from the UCSC genome browser (Nassar et al., 2022), and genome coordinates
264 are from hg38.

B. Relative expression of multiple *CDH4* exon junctions assessed via RT-qPCR after ablating the human-specific CTCF binding motif in human iPSCs or the orthologous region in chimpanzee iPSCs using CRISPR-Cas9. Expression in edited iPSCs is compared to expression in control iPSCs that were subjected to identical nucleofection conditions using a scrambled version of the gRNA. The expected *CDH4* expression pattern is shown in a separate box to the right. Most exon junctions were tested using multiple primers pairs whose name and ID is indicated on the x-axis along with their amplification efficiency, which should ideally fall between 90-110%. The height of the bar plot represents mean *CDH4* expression from N = 3 independent cell culture nucleofection replicates each of which is represented as an individual dot superimposed on the corresponding bar, and error bars represent the SD of these replicates. Note that each cell culture replicate is itself the mean *CDH4* expression of N= 3 technical RT-qPCR reaction replicates.

For all primer pairs, *CDH4* expression is statistically significantly different between both human conditions compared to both chimpanzee conditions. For all primer pairs except those spanning exons 1-2, *CDH4* expression is statistically significantly lower in edited compared to control chimpanzee iPSCs. There is no statistically significant difference in *CDH4* expression between edited and control human iPSCs for any primer pair. Exons are labeled according to the *CDH4* isoform ENST00000614565 as shown in **A**, and p-values were calculated by two-tailed Student's t-test at 5% FDR.

C. Stacked bar chart showing *CDH4* exon usage by the human (left) and chimpanzee (right) alleles in 22 allotetraploid iPSC lines. Of all the RNA-seq reads that map to *CDH4* in a given cell line (Song et al., 2021), each sub-bar represents the proportion that maps (y-axis) to a specific exon (x-axis) after normalizing for length. Exons are labeled according to the *CDH4* transcript ENST00000614565 as shown in **A**.

For our qPCR dosage assay, we designed primers in introns of genes on chromosome 20p (*NDUFAF5*) and chromosome 20q (*CDH4*). For comparison, we also designed primers for genes expected to have the typical copy number of two on chromosomes 3 (*SOX2*), 5 (*MTRR*), 9 (*KLF4*), and 19 (*ATPIA3*), selected because no clonal aberrations were detected on these chromosomes in any of the 33 diploid or tetraploid iPSC lines previously karyotyped by our lab (Song et al., 2021). We confirmed the qPCR assay was working as expected using DNA from a different human diploid cell line (H1) known to contain the iso20q aberration in ~35% of its cells and a different chimpanzee diploid cell line (C1) that does not contain aberrations on any of the chromosomes used for this assay (Song et al., 2021).

We found that all of the cells used in our experiment had similar but unexpected dosage results that are robust to using different standard curves of known DNA concentrations to calculate amplified DNA quantity (Fig. 5.3). Chromosome 20q had higher dosage than chromosome 20p in all of our experimental replicates, which may indicate presence of iso20q. Furthermore, although we expected genes on all other chromosomes to have copy number of two, we instead found chromosomes 9 and 19 had consistently higher dosage than chromosomes 3 and 5, likely indicating significant chromosomal abnormalities beyond iso20q in our experimental replicates. This has a

significant impact on our interpretation of the *CDH4* gene expression levels. Please see the Discussion section for further clarification.

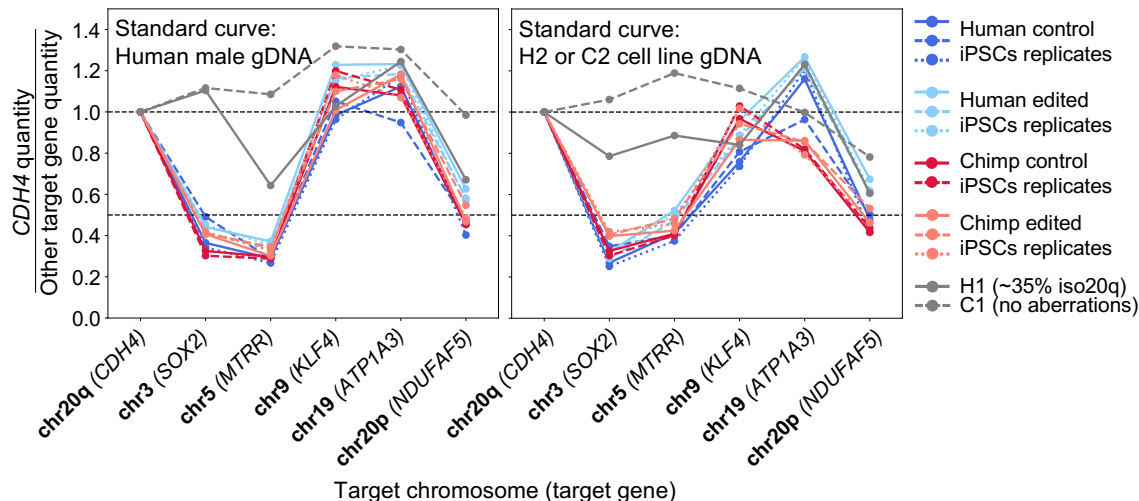


Figure 5.3: Karyotypic aberrations after CRISPR-Cas9 editing

Dosage of chr20q (*CDH4*) relative to other chromosomes assessed via qPCR. Chromosomes used for comparison (and the gene targeted on that chromosome) are on the x-axis, and the DNA quantity ratio of *CDH4* to the other target gene is plotted on the y-axis. Lines connecting data points from the same replicate and dotted horizontal lines at 0.5 and 1 are included to aid visualization.

For each condition, results from $N = 3$ independent cell culture nucleofection replicates are plotted individually and represent the mean value from $N = 3$ technical RT-qPCR reaction replicates. Note that $N = 2$ for chimpanzee control iPSCs due to poor DNA quality from one cell culture replicate following extraction. For comparison, chr20q dosage was also assessed in another human diploid iPSC line (H1) known to contain iso20q at a rate of 35% and another chimpanzee diploid iPSC line (C1) with no known aberrations on the chromosomes included here (Song et al., 2021).

DNA quantity was calculated either by comparing all samples to a standard curve of DNA from a human male (left), or by comparing human samples to a standard curve of DNA from human diploid iPSCs (H2) and comparing chimpanzee samples to a standard curve of DNA from chimpanzee diploid iPSCs (C2). The latter accounts for differences in primer amplification efficiency between species, and results were similar regardless of the standard curve used for quantification.

Discussion

Through our investigation of chromatin arrangement differences between human and chimpanzee iPSCs, we find CTCF binding motifs are enriched in differentially-accessible genomic regions, observe high TAD conservation between species, categorize regions of chromatin

rearrangement by type, demonstrate that differentially expressed genes are more likely to be in regions of dramatic chromatin rearrangement, and begin to characterize an intriguing chromatin rearrangement on chromosome 20 near the gene *CDH4*.

Evolutionary insights from genome-wide chromatin rearrangements between human and chimpanzee iPSCs

That only a small minority of TADs are divergent between our human and chimpanzee iPSCs is largely consistent with previous studies that compare TAD structure in various mammals (Dixon et al., 2012; Vietri Rudan et al., 2015; Krefting et al., 2018; Eres et al., 2019), supporting high TAD structure conservation during mammalian evolution. This implies that dramatic chromatin structural rearrangements are largely selected against during evolution, likely due to their potential to influence the expression of many genes in multiple tissues, potentially leading to deleterious effects. This is further supported by our finding that an even smaller proportion of TADs have severe chromatin rearrangements that completely ablate or create TAD boundaries, probably because modifying the strength or slightly shifting the location of a TAD boundary is less likely to dramatically impact nearby gene expression. Our finding that the number of differentially expressed genes increases with the severity of chromatin rearrangements reinforces previous findings that genes in rearranged TADs tend to show greater gene expression divergence compared to those in conserved TADs across 19 matched tissues in human and mouse (Krefting et al., 2018). Together, these findings suggest that while rare compared to more traditional mechanisms such as modifications to tissue-specific enhancers, TAD rearrangements have occurred during great ape evolution, and may contribute to evolutionarily-important differences in gene regulation.

Furthermore, because we have a panel of human-only, chimpanzee-only, and human-chimpanzee allotetraploid iPSCs, we can differentiate between *cis*-acting and *trans*-acting regulatory changes. *cis* and *trans* regulation is typically discussed in the context of gene expression comparisons between two parental species and an F1 hybrid, where gene expression differences observed between the parental species that are not maintained in the hybrid can be classified as *trans*-acting regulatory differences (Signor and Nuzhdin, 2018). Here, we have the unique opportunity to distinguish between chromatin conformation differences (rather than the typical

gene expression differences) that are due to *cis*- versus *trans*-acting differences. We identified a small set of 9 TADs in the Split, Merge, and Complex categories that are present in single-species but not in allotetraploid comparisons, making them noteworthy candidates for divergence due to *trans* differences. However, manual visual inspection of the TAD structures in these regions reveals they are structurally similar and are likely called in only one dataset because of slight differences in power and thresholding. They are therefore unlikely to represent a biologically-interesting class of species-specific “*trans*” TADs, and it is likely that almost all chromatin arrangement differences between humans and chimpanzees are caused by *cis* differences.

It is also unlikely that differential CTCF binding is driving the majority of the dramatic chromatin rearrangements in our human and chimpanzee iPSCs. Indeed, we identified only one dramatic TAD rearrangement that also has a species-specific CTCF motif located at the species-specific TAD border. Although CTCF is important for establishing TAD borders, ~20% of TAD boundaries remain stable in its absence (Nora et al., 2017), and the dynamic nature of mammalian chromatin structure likely arises from complex interactions between chromatin state and transcriptional activity (Rowley and Corces, 2018). The many dramatic chromatin rearrangements identified between our human and chimpanzee iPSCs that are unaccompanied by sequence changes in CTCF binding sites are noteworthy candidates for future study that could disentangle these interactions.

Insights and challenges after ablating the human-specific CTCF binding motif near *CDH4* with CRISPR-Cas9

The human-specific CTCF binding motif on chromosome 20 near *CDH4*, which we experimentally manipulated using CRISPR-Cas9, is the only example of a known species-specific CTCF binding site at a species-specific TAD border identified in this study. *CDH4* is of particular interest in human evolution due to its potential roles in sensory perception and brain development. In mice, *CDH4* has been associated with enhanced umami perception, making it an intriguing candidate for influencing increased meat consumption in humans compared to the predominantly frugivorous diet of chimpanzees (Zhang et al., 2019). Moreover, *CDH4* influences the radial migration of excitatory projection neurons in the developing neocortical wall (Martinez-Garay et al., 2016), suggesting a potential role in human and chimpanzee cognitive differences.

Still, one of the biggest questions from our work on this chromosome 20 region is why we see such an unexpected *CDH4* expression pattern after ablating the human-specific CTCF binding motif with CRISPR-Cas9 (Fig. 5.2B). Indeed, there are two surprising results: (1) lack of expected *CDH4* expression reduction in edited human iPSCs, and (2) unexpected *CDH4* expression reduction in edited chimpanzee iPSCs. For the unexpected human pattern: It is possible that 3D chromatin arrangement is modified but does not affect *CDH4* expression as we expected; or perhaps 3D chromatin arrangement is not affected at all because other nearby CTCF binding sites compensate. For the unexpected chimpanzee pattern: although unlikely, perhaps the edits introduced by CRISPR-Cas9 inadvertently disrupted a *CDH4* enhancer or somehow inhibited RNA polymerase II.

The presence of multiple annotated transcripts further complicates the interpretation of our *CDH4* expression results in both species. Interestingly, the two exons nearest the human-specific CTCF binding site are present in only one of the four annotated transcripts (Fig. 5.2A). We therefore looked for species-differences in the proportion of reads that map to each *CDH4* exon using RNA-sequencing data from 22 of our human-chimpanzee allotetraploid iPS cell lines (Song et al., 2021) (Fig. 5.2C). Although there is an unexpected bias for certain exons possibly introduced during library preparation (Shi et al., 2021), the two exons closest to the human-specific CTCF binding site (labeled as exons 1 and 2 in Fig. 5.2A) may be used more by human than by chimpanzee iPSCs. While this data remains inconclusive, species differences in *CDH4* isoform usage could impact our interpretation of *CDH4* expression patterns in the CRISPR-Cas9 edited cells, and further investigation into differential isoform usage is warranted.

While biological explanations for the unexpected *CDH4* expression patterns are intriguing, there are technical issues with the experiment itself that make our results difficult to interpret and need to be addressed before drawing conclusions about the biology of this region. First, it is possible that the indels and base pair changes resulting from CRISPR-Cas9 caused unexpected downstream effects that are not representative of the true biology of this locus. A more realistic experimental design that recreates the exact base pair difference present in humans and chimpanzees would provide more confidence in our results. Still, the CRISPR-Cas9 lesions from the experiment reported in this Chapter are largely successful: Our CRISPR gRNA is efficient and

the majority of the cells are edited in both species, the majority of the resulting lesions are small and similar in both species, and the majority of the lesions ablate the human-specific CTCF binding motif in human cells.

It follows, therefore, that the karyotypic abnormalities present in all of our experimental replicates are perhaps the most significant technical issue affecting our interpretation of this experiment. All of our experimental replicates likely have a significant proportion of cells with iso20q (Fig. 5.3), so a significant proportion therefore likely contains three copies of *CDH4*, complicating our ability to interpret *CDH4* expression levels after CRISPR-Cas9 editing. Although we expected some cells may gain aberrations on chromosome 20 because it is a commonly-acquired aberration in human stem cell culture (Amps et al., 2011; Vitillo et al., 2023) and we were inducing double-stranded breaks on this chromosome, we were surprised that all our replicates also acquired abnormalities on other chromosomes, which could have complex and difficult to predict downstream effects. Indeed, even unedited replicates that were subjected to nucleofection with only a scrambled version of the gRNA show chromosomal abnormalities nearly identical to edited replicates that were subjected to nucleofection with the gRNA for the target region. This suggests either that the scrambled gRNA produced lesions despite our efforts to ensure it would not, or that recovering from nucleofection itself was enough to induce significant chromosomal abnormalities. Please note that we are fairly confident our cells had no chromosomal aberrations before nucleofection (Song et al., 2021), although it is possible some abnormalities were gained during normal passaging since they were last karyotyped.

While nucleofection and CRISPR-Cas9 editing can certainly induce significant karyotypic abnormalities in our iPSCs, our previous work with these cells shows it is possible to establish karyotypically normal or nearly-normal stable cell lines after significant genetic manipulation (Song et al., 2021). Given the promising characteristics of this region—its many human-specific features suggesting functional and evolutionary significance, its location near the potentially evolutionarily-interesting gene *CDH4*, and its straightforward single base pair difference between humans and chimpanzees—further study and experimental improvements are certainly warranted for this interesting genomic region.

Materials and Methods

Cell culture maintenance

The induced pluripotent stem cell lines H1 (H23555), H2 (H20961), C1 (C3649), and C2 (C8861) were provided by the Gilad laboratory (Gallego Romero et al., 2015). Tetraploid iPSC lines are from Song *et al.* 2021. iPSCs were routinely propagated feeder-free in mTeSR1 or mTeSR Plus media (STEMCELL Technologies, cat #85850 and cat #100-0276) on cell culture plastics coated with Geltrex basement membrane matrix (Gibco, cat #A1413302). When confluent, cells were passaged using Accutase (Millipore, cat #SCR005) and plated with media supplemented with 1 μ M thiazovivin (Tocris, cat #3845) or using 0.5 mM EDTA as previously described (Beers et al., 2012; Loh et al., 2016). Cells were imaged on the EVOS FL microscope (Thermo Fisher) at 4X magnification.

ATAC-sequencing

ATAC-sequencing was conducted as part of a larger SHARE-seq experiment (Kim et al., 2023) to simultaneously generate single-cell RNA-seq and single-cell ATAC-seq. The samples used were H1C1a allotetraploid iPSCs that were either untreated, treated with camptothecin, or nucleofected with Cas9 and gRNAs targeting a region either on chromosome 21 or X (either with or without ML216). This SHARE-seq experiment was originally part of our effort to induce mitotic recombination between the human and chimpanzee chromosomes in our allotetraploid cells, but because interspecific recombination frequency was low, this data was not included in Song, Grant, Behrens *et al.* 2021. Therefore, for the purposes of this Chapter, we analyzed the resulting ATAC-seq data as if it were bulk, not single-cell, data.

10 SHARE-seq libraries were prepared as previously described (Kim et al., 2023) from iPSC cell pellets flash frozen in liquid nitrogen. Illumina sequencing produced ~22 million ATAC-seq reads (2x150 bp) per library for a total of ~220 million reads. Overall read quality was assessed with FastQC v0.11.9 (Andrews et al., 2010), adapters and low quality bases were removed with cutadapt v3.4 (Martin, 2011), and reads were mapped to a composite human-chimpanzee genome (hg38 and pt6) using STAR 2.7.8a with the flags for STARsolo (Kaminow et al., 2021). Duplicates and unmapped reads were removed with GATK MarkDuplicates (McKenna et al., 2010; Van der

Auwera et al., 2013). ATAC-peaks were called using MAC2 callpeak (Zhang et al., 2008) with sequencing data from haplotagging on the same cells (Song et al., 2021) as the genomic background. Chimpanzee peaks were converted to hg38 coordinates using UCSC's liftOver tool (Nassar et al., 2022), and peaks in ENCODE blacklist genomic regions were removed (Amemiya et al., 2019). A final set of ATAC peaks was generated by retaining only peaks with $-\log_{10}(\text{q-value}) > 3$, which corresponds to peaks with $q < 0.001$, FDR of 5%.

Species-differentially accessible ATAC peaks were called between human peaks and chimpanzee peaks converted to hg38 coordinates using the DESeq2 DiffBind package (Stark and Brown, 2011). A final species-differential peak set was generated by retaining only peaks with corrected p-value < 0.05 . There were no substantial differences when species-differential peaks were called between chimpanzee peaks and human peaks converted to pt6 coordinates, and the results reported in this Chapter are in hg38 coordinates.

Transcription factor binding motif enrichment analysis

Each species-differentially accessible ATAC peak was assigned to the gene it is most likely to regulate using GREAT (McLean et al., 2010). Only those predicted to regulate genes differentially expressed at least in part due to *cis*-regulatory differences as determined in Song, Grant, Behrens *et al.* 2021 were included in the analysis. The genomic sequence of each differentially accessible peak in both hg38 and pt6 coordinates was obtained using pslMap (Zhu et al., 2007) to convert to pt6 coordinates followed by BEDTools getFasta (Quinlan and Hall, 2010). HOMER's findMotifs.pl (Heinz et al., 2010) was used to test for transcription factor binding motif enrichment in species-differentially accessible ATAC peaks; target genomic sequences were from the species in which the ATAC peak is more accessible, and background genomic sequences were the orthologous region in the opposite species where that ATAC peak is less accessible. The location of CTCF motifs within species-differential ATAC peaks was determined using the -find flag with findMotifs.pl.

Hi-C

Hi-C libraries were prepared by Arima Genomics using the Arima-HiC Kit (Arima Genomics, cat#A510008). Libraries were generated from cell pellets that were flash-frozen in

liquid nitrogen for 12 iPS cell lines (Song et al., 2021): two human diploids (H1, H2), two chimpanzee diploids (C1, C2), two human autotetraploids (H1H1a, H1H1b), two chimpanzee autotetraploids (C1C1b, C1C1e), and four human-chimpanzee allotetraploids (H1C1h, H1H1k, H2C2a, H2C2d)

For the eight diploid and autotetraploid cell lines, one library per cell line was prepared and sequenced to a depth of ~800 million reads (2x150 bp). For the four allotetraploid cell lines, two libraries per cell line were prepared and sequenced to a depth of ~1.1 billion reads (2x150 bp) each; the reads from the two libraries were combined for downstream analyses, resulting in a total of ~2.2 billion reads for each of the four allotetraploid cell lines. The prepared libraries were sequenced on an Illumina NovaSeq 6000.

Overall read quality was assessed using FastQC v0.11.9 (Andrews et al., 2010), and adapters and low-quality bases were removed with cutadapt v3.4 (Martin, 2011). R1 and R2 reads were separately mapped to a composite human-chimpanzee genome (hg38 and pt6) using BWA-MEM (Li, 2013). Only read pairs where at least one of R1 or R2 mapped uniquely and where neither mapped to an ENCODE blacklisted region (Amemiya et al., 2019) were retained.

Retained pairs were processed through the Juicer pipeline (Durand et al., 2016), which included converting aligned reads to Juicer format using the pre script, removing chimeric reads and duplicates, and generating contact matrices at 1 kb, 5 kb, and 10 kb resolutions. Contact matrices were normalized using the iterative correction and eigenvector decomposition (ICE) method integrated into Juicer, and the 5 kb resolution maps were used for downstream analyses. Juicebox (Robinson et al., 2018) and Hi-Glass (Kerpedjiev et al., 2018) were used for data visualization.

Using ConsensusTAD (Cresswell and Dozmorov, 2020), a set of consensus TAD boundaries that were consistent across similar cell lines was generated for each of the following: human-only cell lines, chimpanzee-only cell lines, the human chromosome from human-chimpanzee cell lines, and the chimpanzee chromosome from human-chimpanzee cell lines. The consensus TAD boundaries were used for analysis with TADCompare (Cresswell and Dozmorov, 2020), which identified species-differential TAD boundaries and categorized species-specific rearrangements as “Split,” “Merge,” “Complex,” “Shifted,” or “Strength Change” (Table 5.2).

Note that only boundaries called by both the HiCEXplorer (Ramírez et al., 2018) and TADtree (Weinreb and Raphael, 2016) algorithms in at least one cell line were considered by TADcompare, as specified with the “pre_tads” variable.

CRISPR-Cas9 editing in iPSCs

CRISPR-Cas9 gRNAs were designed using CHOPCHOP (Labun et al., 2019) and *in vitro* transcribed (IVT) as previously described (Wucherpfennig et al., 2019). Briefly, CRISPR IVT target oligos containing the gRNA and the CRISPR IVT scaffold oligo (5'-GATCCGCACCGACTCGGTGCCACTTTTTCAAGTTGATAACGGACTAGCCTTATTTAACTTGCTATTTCTAGCTCTAAAAC-3', HPLC-purified) were synthesized by Integrated DNA Technologies. 40 PCR cycles were performed between the CRISPR IVT scaffold oligo and the CRISPR IVT target oligo using Phusion DNA polymerase (Thermo Scientific, cat #F530L), and the PCR product was purified using the QIAquick PCR Purification Kit (Qiagen, cat #28104). The PCR product was then *in vitro* transcribed using the MEGAscript T7 transcription kit (Thermo Fisher, cat #AM1334) for 16 hours at 37°C. The reaction was treated with DNase, and transcribed gRNAs were extracted with LiCl precipitation. Final transcribed gRNAs were resuspended to approximately 2 µg/ul.

Cas9 and gRNAs were delivered to iPSCs via nucleofection in 96-well plate format. 1 µl of 40 µM Cas9-NLS purified protein (QB3, UC Berkeley) was complexed with 2 µg of gRNA for 10 minutes at room temperature per reaction. 2×10^5 cells per reaction were nucleofected using the P3 Primary Cell 96-well Nucleofector Kit (Lonza, cat #V4SP-3096) on the Amaxa 96-well Shuttle Device (Lonza) with program CA-137. Cells were plated in media supplemented with CloneR™2 (STEMCELL Technologies, cat #100-069) post-nucleofection and allowed to recover for two days before collection.

To select the highest efficiency gRNA, both human diploid and chimpanzee diploid iPSCs were nucleofected with each of 6 gRNAs. DNA from nucleofected cells was extracted using the QIAamp DNA Micro Kit (Qiagen, cat #56304) and PCR amplified using primers bracketing the cutsite (forward primer: 5'-AGAAACCAGGACCTCAGCAC-3' / reverse primer: 5'-AGCCATGGCCTTCTCAACA-3'). The PCR product was purified with the MinElute PCR Purification Kit (Qiagen, cat #28004) and sent for Sanger sequencing. Resulting traces were analyzed using ICE analysis (Synthego Performance Analysis, 2019, v3.0) to estimate guide

efficiency. Only the most efficient gRNA (CRISPR IVT target oligo 5'-AATTAATACGACTCACTATAGAAAGGACCTTCTCCACAAGGGTTTTAGAGCTAGAAATAGC-3') was used for downstream experiments.

To ablate the human-specific CTCF binding motif, three replicates of human diploid or chimpanzee diploid cells were nucleofected either with the most efficient gRNA (as determined above), or with a scrambled gRNA that does not target any locus in either the human or the chimpanzee genome (CRISPR IVT scrambled oligo: 5'-AATTAATACGACTCACTATAGCGAAATCCGTGCCCCGAATAGTTTTAGAGCTAGAAATAGC-3'). Each cell culture replicate was composed of 4 nucleofection reactions that were pooled post-nucleofection and plated into one well of a 24-well plate. DNA and RNA from nucleofected cells was co-extracted with the Quick-DNA/RNA Microprep Plus kit (Zymo, cat# D7005). To assess cutting efficiency and examine individual resulting lesions, DNA was PCR amplified using primers bracketing the cutsite (forward primer: 5'-TTGTCATGCCCATTAATCCCC-3' / reverse primer: 5'-CTCATTTCTGGGGCCGTCTA-3') and sent for Amplicon-EZ sequencing (2x250 bp) at Azenta Life Sciences. Resulting reads were checked for quality using FastQC (Andrews et al., 2010), and adapters and nucleotides of poor quality were removed with Trimmomatic version 0.36 (Bolger et al., 2014). If the forward and reverse reads overlapped, paired sequence reads were merged to form a single read. These merged reads were then aligned to a reference sequence matching the expected PCR product, and variant detection was performed using Azenta's Amplicon-EZ program.

Quantitative PCR (qPCR)

RT-qPCR was used to assess *CDH4* expression, and qPCR was used to assess chr20 dosage in cells post-nucleofection. All reactions were performed in triplicate using Brilliant II SYBR Green Low ROX qPCR Master Mix (Agilent, cat #600830) on a QuantStudio 5 Real-Time PCR System (Thermo Fisher). For RT-qPCR, cDNA was synthesized from RNA with the SuperScriptTM VILOTM cDNA Synthesis Kit (Thermo Fisher, cat #11754250).

Absolute input quantity of cDNA (RT-qPCR) or gDNA (qPCR) was calculated using a standard curve of cDNA pooled from all samples (RT-qPCR), or using a standard curve of gDNA from a human male (Promega, cat #G1471), human diploid iPSCs, or chimpanzee diploid iPSCs (qPCR). This was done by plotting the known quantities of the standard against their corresponding

quantification cycle (Cq) values to generate a linear regression equation (quantity = slope * Cq + intercept). The Cq values from the experimental samples were then used in this equation to determine their absolute quantities.

These absolute quantities were then normalized. For RT-qPCR, *CDH4* quantity was normalized by dividing by the geometric mean of the quantities of two housekeeping genes (*GAPDH*, *YWHAZ*) in the same sample, and then divided by the normalized quantity of *CDH4* in unedited human cells. two-tailed Student's t-tests were used to determine statistically significant differences in *CDH4* expression between experimental conditions at 5% Benjamini-Hochberg FDR. For qPCR, relative chr20q dosage was calculated by dividing *CDH4* quantity by the quantity of genes on other chromosomes. All primer pairs bind identical sequences in humans and chimpanzees and produce PCR products of identical length in both species. For RT-qPCR, primer pairs span various exon junctions in *CDH4*. For qPCR, primer pairs are within the intron of the target gene with the exception of *SOX2*, which lacks introns; in this case, primers are downstream.

RT-qPCR primers (ID: Forward primer / Reverse primer):

- ***CDH4* Exons 1-2 (RT14):**
5'-CTGCTGCTCTCGCTCTC-3' / 5'-AATGCCGTGTAATCATCTTCAG-3'
- ***CDH4* Exons 1-2 (RT15):**
5'-CTGCTGCTCTCGCTCTC-3' / 5'-TGGGAGATTAATGCCGTGTAA-3'
- ***CDH4* Exons 2-3 (RT16):**
5'-GGCTGGGTTCTCTGAAGAT-3' / 5'-CCATGCTGTTGGTCTCATATTG-3'
- ***CDH4* Exons 2-3 (RT17):**
5'-TGGGTTCTCTGAAGATGATTACA-3' / 5'-ACTTTGAAGTCCATGCTGTTG-3'
- ***CDH4* Exons 2-3 (RT18):**
5'-CTGAAGATGATTACACGGCATT-3' / 5'-CCACACAGCTGCTGAAC-3'
- ***CDH4* Exons 2-3 (RT19):**
5'-TGCAAGGCTGGGTTCTC-3' / 5'-CTGAACTTGACTTGAAGTAGCTTT-3'
- ***CDH4* Exons 4-5 (RT02):**
5'-CATCAACGTGCCCCGAGAACT-3' / 5'-GTCAATGCTGAAGACCTCCAT-3'
- ***CDH4* Exons 8-9 (RT05):**
5'-TGACAGATGTGAATGACAACCC-3' / 5'-GGCATTCCAGTTTGGAGAGT-3'
- ***CDH4* Exons 8-9 (RT06):**
5'-ACACAGCCACAGCCATCAT-3' / 5'-CGGTCTCCACGCGGTTT-3'
- ***CDH4* Exons 8-9 (RT07):**
5'-CATCACGGTGACAGATGTGAA-3' / 5'-GGCATTCCAGTTTGGAGAGT-3'

- 615 • ***CDH4* Exons 8-9 (RT08):**
616 5'-GCCATCATCACGGTGACAG-3' / 5'-GCTGATCTCGGTCCATCAC-3'
- 617 • ***GAPDH*:**
618 5'-GGTCGGAGTCAACGGATT-3' / 5'-GATGGCAACAATATCCACTTTACC-3'
- 619 • ***YWHAZ*:**
620 5'-ACAAAGACAGCACGCTAATAATG-3' / 5'-GCTTCAGCTTCGTCTCCTT-3'
- 621 RT-qPCR primers (ID: Forward primer / Reverse primer):
- 622 • **chr3, *SOX2* (33):**
623 5'-TATTGTAAATCCGCCCCCTCCT-3' / 5'-ATTGAATCCTTTCTGGCAACAAGT-3'
- 624 • **chr5, *MTRR* (23):**
625 5'-TTGCCGTGCGTGTTAGAGTT-3' / 5'-CTAGTCCCTGAAGAAGCAGCC-3'
- 626 • **chr9, *KLF4* (34):**
627 5'-ACCAAGAGCCACTGAACGAG-3' / 5'-GCCAAGGACACGGAAGCTAT-3'
- 628 • **chr19, *ATPLA3* (28):**
629 5'-GCACTGCCTGTCCCATTCTT-3' / 5'-GCTACCTCTGTGGCTGCAAT-3'
- 630 • **chr20p, *NDUFAF5* (29):**
631 5'-GCTCTTAACTGTCTTGCCCCCT-3' / 5'-ACCACTGATGCTCCCCAAAT-3'
- 632 • **chr20q, *CDH4* (06):**
633 5'-CCTCACACCCTCTCTAGGCT-3' / 5'-TTCAGAGGCTCTTTCTGCCG-3'

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References

- Amemiya, H.M., Kundaje, A., Boyle, A.P., 2019. The ENCODE blacklist: Identification of problematic regions of the genome. *Scientific Reports* 9, 9354.
<https://doi.org/10.1038/s41598-019-45839-z>
- Amps, K., Andrews, P.W., Anyfantis, G., Armstrong, L., Avery, S., Baharvand, H., Baker, J., Baker, D., Munoz, M.B., Beil, S., Benvenisty, N., Ben-Yosef, D., Biancotti, J.-C., Bosman, A., Brena, R.M., Brison, D., Caisander, G., Camarasa, M.V., Chen, J., Chiao, E., Choi, Y.M., Choo, A.B.H., Collins, D., Colman, A., Crook, J.M., Daley, G.Q., Dalton, A., De Sousa, P.A., Denning, C., Downie, J., Dvorak, P., Montgomery, K.D., Feki, A., Ford, A., Fox, V., Fraga, A.M., Frumkin, T., Ge, L., Gokhale, P.J., Golan-Lev, T., Gourabi, H., Gropp, M., Guangxiu, L., Hampl, A., Harron, K., Healy, L., Herath, W., Holm, F., Hovatta, O., Hyllner, J., Inamdar, M.S., Irwanto, A.K., Ishii, T., Jaconi, M., Jin, Y., Kimber, S., Kiselev, S., Knowles, B.B., Kopper, O., Kukhareenko, V., Kuliev, A., Lagarkova, M.A., Laird, P.W., Lako, M., Laslett, A.L., Lavon, N., Lee, D.R., Lee, J.E., Li, C., Lim, L.S., Ludwig, T.E., Ma, Y., Maltby, E., Mateizel, I., Mayshar, Y., Mileikovsky, M., Minger, S.L., Miyazaki, T., Moon, S.Y., Moore, H., Mummery, C., Nagy, A., Nakatsuji, N., Narwani, K., Oh, S.K.W., Oh, S.K., Olson, C., Otonkoski, T., Pan, F., Park, I.-H., Pells, S., Pera, M.F., Pereira, L.V., Qi, O., Raj, G.S., Reubinoff, B., Robins, A., Robson, P., Rossant, J., Salekdeh, G.H., Schulz, T.C., Sermon, K., Mohamed, J.S., Shen, H., Sherrer, E., Sidhu, K., Sivarajah, S., Skottman, H., Spits, C., Stacey, G.N., Strehl, R., Strelchenko, N., Suemori, H., Sun, B., Suuronen, R., Takahashi, K., Tuuri, T., Venu, P., Verlinsky, Y., Oostwaard, D.W., Weisenberger, D.J., Wu, Y., Yamanaka, S., Young, L., Zhou, Q., The International Stem Cell Initiative, 2011. Screening ethnically diverse human embryonic stem cells identifies a chromosome 20 minimal amplicon conferring growth advantage. *Nature Biotechnology* 29, 1132–1144.
<https://doi.org/10.1038/nbt.2051>
- Andrews, S., Krueger, F., Segonds-Pichon, A., Biggins, L., Krueger, C., Wingett, S., 2010. FastQC: A quality control tool for high throughput sequence data.
<http://www.bioinformatics.babraham.ac.uk/projects/fastqc>
- Beers, J., Gulbranson, D.R., George, N., Siniscalchi, L.I., Jones, J., Thomson, J.A., Chen, G., 2012. Passaging and colony expansion of human pluripotent stem cells by enzyme-free dissociation in chemically defined culture conditions. *Nature Protocols* 7, 2029–2040.
<https://doi.org/10.1038/nprot.2012.130>
- Bolger, A.M., Lohse, M., Usadel, B., 2014. Trimmomatic: A flexible trimmer for Illumina sequence data. *Bioinformatics* 30, 2114–2120.
<https://doi.org/10.1093/bioinformatics/btu170>
- Carroll, S.B., 2008. Evo-Devo and an expanding evolutionary synthesis: A genetic theory of morphological evolution. *Cell* 134, 25–36. <https://doi.org/10.1016/j.cell.2008.06.030>

679 Carroll, S.B., 2005. Evolution at two levels: On genes and form. *PLoS Biology* 3, e245.
680 <https://doi.org/10.1371/journal.pbio.0030245>

681 Cresswell, K.G., Dozmorov, M.G., 2020. TADCompare: An R package for differential and
682 temporal analysis of topologically associated domains. *Frontiers in Genetics* 11.
683 <https://doi.org/10.3389/fgene.2020.00158>

684 Dekker, J., Heard, E., 2015. Structural and functional diversity of topologically associating
685 domains. *FEBS Letters* 589, 2877–2884. <https://doi.org/10.1016/j.febslet.2015.08.044>

686 Dixon, J.R., Selvaraj, S., Yue, F., Kim, A., Li, Y., Shen, Y., Hu, M., Liu, J.S., Ren, B., 2012.
687 Topological domains in mammalian genomes identified by analysis of chromatin
688 interactions. *Nature* 485, 376–380. <https://doi.org/10.1038/nature11082>

689 Dougherty, C.J., Ichim, T.E., Liu, L., Reznik, G., Min, W.-P., Ghochikyan, A., Agadjanyan,
690 M.G., Reznik, B.N., 2008. Selective apoptosis of breast cancer cells by siRNA targeting
691 of BORIS. *Biochemical and Biophysical Research Communications* 370, 109–112.
692 <https://doi.org/10.1016/j.bbrc.2008.03.040>

693 Durand, N.C., Shamim, M.S., Machol, I., Rao, S.S.P., Huntley, M.H., Lander, E.S., Aiden, E.L.,
694 2016. Juicer provides a one-click system for analyzing loop-resolution Hi-C experiments.
695 *Cell Systems* 3, 95–98. <https://doi.org/10.1016/j.cels.2016.07.002>

696 Eres, I.E., Luo, K., Hsiao, C.J., Blake, L.E., Gilad, Y., 2019. Reorganization of 3D genome
697 structure may contribute to gene regulatory evolution in primates. *PLoS Genetics* 15,
698 e1008278. <https://doi.org/10.1371/journal.pgen.1008278>

699 Franke, M., Ibrahim, D.M., Andrey, G., Schwarzer, W., Heinrich, V., Schöpflin, R., Kraft, K.,
700 Kempfer, R., Jerković, I., Chan, W.-L., Spielmann, M., Timmermann, B., Wittler, L.,
701 Kurth, I., Cambiaso, P., Zuffardi, O., Houge, G., Lambie, L., Brancati, F., Pombo, A.,
702 Vingron, M., Spitz, F., Mundlos, S., 2016. Formation of new chromatin domains
703 determines pathogenicity of genomic duplications. *Nature* 538, 265–269.
704 <https://doi.org/10.1038/nature19800>

705 Fudenberg, G., Pollard, K.S., 2019. Chromatin features constrain structural variation across
706 evolutionary timescales. *Proceedings of the National Academy of Sciences USA* 116,
707 2175–2180. <https://doi.org/10.1073/pnas.1808631116>

708 Gallego Romero, I., Pavlovic, B.J., Hernando-Herraez, I., Zhou, X., Ward, M.C., Banovich,
709 N.E., Kagan, C.L., Burnett, J.E., Huang, C.H., Mitrano, A., Chavarria, C.I., Friedrich
710 Ben-Nun, I., Li, Y., Sabatini, K., Leonardo, T.R., Parast, M., Marques-Bonet, T., Laurent,
711 L.C., Loring, J.F., Gilad, Y., 2015. A panel of induced pluripotent stem cells from
712 chimpanzees: A resource for comparative functional genomics. *eLife* 4, e07103.
713 <https://doi.org/10.7554/eLife.07103>

714 Giorgio, E., Robyr, D., Spielmann, M., Ferrero, E., Di Gregorio, E., Imperiale, D., Vaula, G.,
715 Stamoulis, G., Santoni, F., Atzori, C., Gasparini, L., Ferrera, D., Canale, C., Guipponi,
716 M., Pennacchio, L.A., Antonarakis, S.E., Brussino, A., Brusco, A., 2015. A large

- genomic deletion leads to enhancer adoption by the *lamin B1* gene: a second path to autosomal dominant adult-onset demyelinating leukodystrophy (ADLD). *Human Molecular Genetics* 24, 3143–3154. <https://doi.org/10.1093/hmg/ddv065>
- Gompel, N., Prud'homme, B., Wittkopp, P.J., Kassner, V.A., Carroll, S.B., 2005. Chance caught on the wing: *cis*-regulatory evolution and the origin of pigment patterns in *Drosophila*. *Nature* 433, 481–487. <https://doi.org/10.1038/nature03235>
- Heinz, S., Benner, C., Spann, N., Bertolino, E., Lin, Y.C., Laslo, P., Cheng, J.X., Murre, C., Singh, H., Glass, C.K., 2010. Simple combinations of lineage-determining transcription factors prime *cis*-regulatory elements required for macrophage and B cell identities. *Molecular Cell* 38, 576–589. <https://doi.org/10.1016/j.molcel.2010.05.004>
- Hnisz, D., Weintraub, A.S., Day, D.S., Valton, A.-L., Bak, R.O., Li, C.H., Goldmann, J., Lajoie, B.R., Fan, Z.P., Sigova, A.A., Reddy, J., Borges-Rivera, D., Lee, T.I., Jaenisch, R., Porteus, M.H., Dekker, J., Young, R.A., 2016. Activation of proto-oncogenes by disruption of chromosome neighborhoods. *Science* 351, 1454–1458. <https://doi.org/10.1126/science.aad9024>
- Hubisz, M.J., Pollard, K.S., 2014. Exploring the genesis and functions of Human Accelerated Regions sheds light on their role in human evolution. *Current Opinion in Genetics & Development* 29, 15–21. <https://doi.org/10.1016/j.gde.2014.07.005>
- Kaminow, B., Yunusov, D., Dobin, A., 2021. STARsolo: Accurate, fast and versatile mapping/quantification of single-cell and single-nucleus RNA-seq data. *bioRxiv*. <https://doi.org/10.1101/2021.05.05.442755>
- Kim, S.H., Marinov, G.K., Bagdatli, S.T., Higashino, S.I., Shipony, Z., Kundaje, A., Greenleaf, W.J., 2023. Simultaneous single-cell profiling of the transcriptome and accessible chromatin using SHARE-seq. *Methods in Molecular Biology* 2611, 187–230. https://doi.org/10.1007/978-1-0716-2899-7_11
- King, M.-C., Wilson, A.C., 1975. Evolution at two levels in humans and chimpanzees. *Science* 188, 107–116. <https://doi.org/10.1126/science.1090005>
- Krefting, J., Andrade-Navarro, M.A., Ibn-Salem, J., 2018. Evolutionary stability of topologically associating domains is associated with conserved gene regulation. *BMC Biology* 16, 1–12. <https://doi.org/10.1186/s12915-018-0556-x>
- Labun, K., Montague, T.G., Krause, M., Torres Cleuren, Y.N., Tjeldnes, H., Valen, E., 2019. CHOPCHOP v3: Expanding the CRISPR web toolbox beyond genome editing. *Nucleic Acids Research* 47, W171–W174. <https://doi.org/10.1093/nar/gkz365>
- Landrum, M.J., Lee, J.M., Benson, M., Brown, G.R., Chao, C., Chitpiralla, S., Gu, B., Hart, J., Hoffman, D., Jang, W., Karapetyan, K., Katz, K., Liu, C., Maddipatla, Z., Malheiro, A., McDaniel, K., Ovetsky, M., Riley, G., Zhou, G., Holmes, J.B., Kattman, B.L., Maglott, D.R., 2018. ClinVar: Improving access to variant interpretations and supporting evidence. *Nucleic Acids Research* 46, D1062–D1067.

755 <https://doi.org/10.1093/nar/gkx1153>

756 Lazar, N.H., Nevonen, K.A., O’Connell, B., McCann, C., O’Neill, R.J., Green, R.E., Meyer, T.J.,
 757 Okhovat, M., Carbone, L., 2018. Epigenetic maintenance of topological domains in the
 758 highly rearranged gibbon genome. *Genome Research* 28, 983–997.
 759 <https://doi.org/10.1101/gr.233874.117>

760 Levine, M., Tjian, R., 2003. Transcription regulation and animal diversity. *Nature* 424, 147–151.
 761 <https://doi.org/10.1038/nature01763>

762 Li, H., 2013. Aligning sequence reads, clone sequences and assembly contigs with BWA-MEM.
 763 *arXiv*. <https://arxiv.org/abs/1303.3997v2>

764 Liu, Q., Chen, K., Liu, Z., Huang, Y., Zhao, R., Wei, L., Yu, X., He, J., Liu, J., Qi, J., Qin, Y.,
 765 Li, B., 2017. BORIS up-regulates OCT4 via histone methylation to promote cancer stem
 766 cell-like properties in human liver cancer cells. *Cancer Letters* 403, 165–174.
 767 <https://doi.org/10.1016/j.canlet.2017.06.017>

768 Loh, K.M., Chen, A., Koh, P.W., Deng, T.Z., Sinha, R., Tsai, J.M., Barkal, A.A., Shen, K.Y.,
 769 Jain, R., Morganti, R.M., Shyh-Chang, N., Fernhoff, N.B., George, B.M., Wernig, G.,
 770 Salomon, R.E.A., Chen, Z., Vogel, H., Epstein, J.A., Kundaje, A., Talbot, W.S., Beachy,
 771 P.A., Ang, L.T., Weissman, I.L., 2016. Mapping the pairwise choices leading from
 772 pluripotency to human bone, heart, and other mesoderm cell types. *Cell* 166, 451–467.
 773 <https://doi.org/10.1016/j.cell.2016.06.011>

774 Loukinov, D.I., Pugacheva, E., Vatolin, S., Pack, S.D., Moon, H., Chernukhin, I., Mannan, P.,
 775 Larsson, E., Kanduri, C., Vostrov, A.A., Cui, H., Niemitz, E.L., Rasko, J.E.J., Docquier,
 776 F.M., Kistler, M., Breen, J.J., Zhuang, Z., Quitschke, W.W., Renkawitz, R., Klenova,
 777 E.M., Feinberg, A.P., Ohlsson, R., Morse, H.C., Lobanenko, V.V., 2002. BORIS, a
 778 novel male germ-line-specific protein associated with epigenetic reprogramming events,
 779 shares the same 11-zinc-finger domain with CTCF, the insulator protein involved in
 780 reading imprinting marks in the soma. *Proceedings of the National Academy of Sciences*
 781 *USA* 99, 6806–6811. <https://doi.org/10.1073/pnas.092123699>

782 Lupiáñez, D.G., Kraft, K., Heinrich, V., Krawitz, P., Brancati, F., Klopocki, E., Horn, D.,
 783 Kayserili, H., Opitz, J.M., Laxova, R., Santos-Simarro, F., Gilbert-Dussardier, B.,
 784 Wittler, L., Borschiwer, M., Haas, S.A., Osterwalder, M., Franke, M., Timmermann, B.,
 785 Hecht, J., Spielmann, M., Visel, A., Mundlos, S., 2015. Disruptions of topological
 786 chromatin domains cause pathogenic rewiring of gene-enhancer interactions. *Cell* 161,
 787 1012–1025. <https://doi.org/10.1016/j.cell.2015.04.004>

788 Lupiáñez, D.G., Spielmann, M., Mundlos, S., 2016. Breaking TADs: How alterations of
 789 chromatin domains result in disease. *Trends in Genetics* 32, 225–237.
 790 <https://doi.org/10.1016/j.tig.2016.01.003>

791 Martin, M., 2011. Cutadapt removes adapter sequences from high-throughput sequencing reads.
 792 *EMBnet Journal* 17, 10–12. <http://dx.doi.org/10.14806/ej.17.1.200>

793 Martinez-Garay, I., Gil-Sanz, C., Franco, S.J., Espinosa, A., Molnár, Z., Mueller, U., 2016.
794 Cadherin 2/4 signaling via PTP1B and catenins is crucial for nucleokinesis during radial
795 neuronal migration in the neocortex. *Development* 143, 2121–2134.
796 <https://doi.org/10.1242/dev.132456>

797 McKenna, A., Hanna, M., Banks, E., Sivachenko, A., Cibulskis, K., Kernytsky, A., Garimella,
798 K., Altshuler, D., Gabriel, S., Daly, M., DePristo, M.A., 2010. The Genome Analysis
799 Toolkit: A MapReduce framework for analyzing next-generation DNA sequencing data.
800 *Genome Research* 20, 1297–1303. <https://doi.org/10.1101/gr.107524.110>

801 McLean, C.Y., Bristor, D., Hiller, M., Clarke, S.L., Schaar, B.T., Lowe, C.B., Wenger, A.M.,
802 Bejerano, G., 2010. GREAT improves functional interpretation of *cis*-regulatory regions.
803 *Nature Biotechnology* 28, 495–501. <https://doi.org/10.1038/nbt.1630>

804 McLean, C.Y., Reno, P.L., Pollen, A.A., Bassan, A.I., Capellini, T.D., Guenther, C., Indjeian,
805 V.B., Lim, X., Menke, D.B., Schaar, B.T., Wenger, A.M., Bejerano, G., Kingsley, D.M.,
806 2011. Human-specific loss of regulatory DNA and the evolution of human-specific traits.
807 *Nature* 471, 216–219. <https://doi.org/10.1038/nature09774>

808 Nassar, L.R., Barber, G.P., Benet-Pagès, A., Casper, J., Clawson, H., Diekhans, M., Fischer, C.,
809 Gonzalez, J.N., Hinrichs, A.S., Lee, B.T., Lee, C.M., Muthuraman, P., Nguy, B., Pereira,
810 T., Nejad, P., Perez, G., Raney, B.J., Schmelter, D., Speir, M.L., Wick, B.D., Zweig,
811 A.S., Haussler, D., Kuhn, R.M., Haeussler, M., Kent, W.J., 2022. The UCSC Genome
812 Browser database: 2023 update. *Nucleic Acids Research* 51, D1188–D1195.
813 <https://doi.org/10.1093/nar/gkac1072>

814 Nora, E.P., Goloborodko, A., Valton, A.-L., Gibcus, J.H., Uebersohn, A., Abdennur, N., Dekker,
815 J., Mirny, L.A., Bruneau, B.G., 2017. Targeted degradation of CTCF decouples local
816 insulation of chromosome domains from genomic compartmentalization. *Cell* 169, 930-
817 944.e22. <https://doi.org/10.1016/j.cell.2017.05.004>

818 Olds, L.C., Sibley, E., 2003. Lactase persistence DNA variant enhances lactase promoter activity
819 *in vitro*: Functional role as a *cis* regulatory element. *Human Molecular Genetics* 12,
820 2333–2340. <https://doi.org/10.1093/hmg/ddg244>

821 Phillips-Cremins, J.E., Sauria, M.E.G., Sanyal, A., Gerasimova, T.I., Lajoie, B.R., Bell, J.S.K.,
822 Ong, C.-T., Hookway, T.A., Guo, C., Sun, Y., Bland, M.J., Wagstaff, W., Dalton, S.,
823 McDevitt, T.C., Sen, R., Dekker, J., Taylor, J., Corces, V.G., 2013. Architectural protein
824 subclasses shape 3D organization of genomes during lineage commitment. *Cell* 153,
825 1281–1295. <https://doi.org/10.1016/j.cell.2013.04.053>

826 Pollard, K.S., Salama, S.R., Lambert, N., Lambot, M.-A., Coppens, S., Pedersen, J.S., Katzman,
827 S., King, B., Onodera, C., Siepel, A., Kern, A.D., Dehay, C., Igel, H., Ares, M.,
828 Vanderhaeghen, P., Haussler, D., 2006. An RNA gene expressed during cortical
829 development evolved rapidly in humans. *Nature* 443, 167–172.
830 <https://doi.org/10.1038/nature05113>

831 Quinlan, A.R., Hall, I.M., 2010. BEDTools: A flexible suite of utilities for comparing genomic

832 features. *Bioinformatics* 26, 841–842. <https://doi.org/10.1093/bioinformatics/btq033>

833 Rao, S.S.P., Huntley, M.H., Durand, N.C., Stamenova, E.K., Bochkov, I.D., Robinson, J.T.,
834 Sanborn, A.L., Machol, I., Omer, A.D., Lander, E.S., Aiden, E.L., 2014. A 3D map of the
835 human genome at kilobase resolution reveals principles of chromatin looping. *Cell* 159,
836 1665–1680. <https://doi.org/10.1016/j.cell.2014.11.021>

837 Real, F.M., Haas, S.A., Franchini, P., Xiong, P., Simakov, O., Kuhl, H., Schöpflin, R., Heller,
838 D., Moeinzadeh, M.-H., Heinrich, V., Krannich, T., Bressin, A., Hartmann, M.F., Wudy,
839 S.A., Dechmann, D.K.N., Hurtado, A., Barrionuevo, F.J., Schindler, M., Harabula, I.,
840 Osterwalder, M., Hiller, M., Wittler, L., Visel, A., Timmermann, B., Meyer, A., Vingron,
841 M., Jiménez, R., Mundlos, S., Lupiáñez, D.G., 2020. The mole genome reveals
842 regulatory rearrangements associated with adaptive intersexuality. *Science* 370, 208–214.
843 <https://doi.org/10.1126/science.aaz2582>

844 Renaud, S., Loukinov, D., Alberti, L., Vostrov, A., Kwon, Y.-W., Bosman, F.T., Lobanenko,
845 V., Benhattar, J., 2011. BORIS/CTCF-mediated transcriptional regulation of the hTERT
846 telomerase gene in testicular and ovarian tumor cells. *Nucleic Acids Research* 39, 862–
847 873. <https://doi.org/10.1093/nar/gkq827>

848 Robinson, J.T., Turner, D., Durand, N.C., Thorvaldsdóttir, H., Mesirov, J.P., Aiden, E.L., 2018.
849 Juicebox.js provides a cloud-based visualization system for Hi-C data. *Cell Systems* 6,
850 256–258.e1. <https://doi.org/10.1016/j.cels.2018.01.001>

851 Rowley, M.J., Corces, V.G., 2018. Organizational principles of 3D genome architecture. *Nature*
852 *Reviews Genetics* 19, 789–800. <https://doi.org/10.1038/s41576-018-0060-8>

853 Shi, H., Zhou, Y., Jia, E., Pan, M., Bai, Y., Ge, Q., 2021. Bias in RNA-seq library preparation:
854 Current challenges and solutions. *BioMed Research International* 2021, 6647597.
855 <https://doi.org/10.1155/2021/6647597>

856 Signor, S.A., Nuzhdin, S.V., 2018. The evolution of gene expression in *cis* and *trans*. *Trends in*
857 *Genetics* 34, 532–544. <https://doi.org/10.1016/j.tig.2018.03.007>

858 Song, J.H.T., Grant, R.L., Behrens, V.C., Kučka, M., Roberts Kingman, G.A., Soltys, V., Chan,
859 Y.F., Kingsley, D.M., 2021. Genetic studies of human–chimpanzee divergence using
860 stem cell fusions. *Proceedings of the National Academy of Sciences USA* 118,
861 e2117557118. <https://doi.org/10.1073/pnas.2117557118>

862 Stark, R., Brown, G., 2011. DiffBind: Differential binding analysis of ChIP-Seq peak data. *R*
863 *Package Version* 100, 1–40.
864 <http://bioconductor.org/packages/release/bioc/vignettes/DiffBind/inst/doc/DiffBind.pdf>

865 Symmons, O., Uslu, V.V., Tsujimura, T., Ruf, S., Nassari, S., Schwarzer, W., Ettwiller, L., Spitz,
866 F., 2014. Functional and topological characteristics of mammalian regulatory domains.
867 *Genome Research* 24, 390–400. <https://doi.org/10.1101/gr.163519.113>

868 Van Bortle, K., Nichols, M.H., Li, L., Ong, C.-T., Takenaka, N., Qin, Z.S., Corces, V.G., 2014.
869 Insulator function and topological domain border strength scale with architectural protein

870 occupancy. *Genome Biology* 15, R82. <https://doi.org/10.1186/gb-2014-15-5-r82>

871 Van der Auwera, G.A., Carneiro, M.O., Hartl, C., Poplin, R., Del Angel, G., Levy-Moonshine,
872 A., Jordan, T., Shakir, K., Roazen, D., Thibault, J., Banks, E., Garimella, K.V., Altshuler,
873 D., Gabriel, S., DePristo, M.A., 2013. From FastQ data to high confidence variant calls:
874 The Genome Analysis Toolkit best practices pipeline. *Current Protocols in*
875 *Bioinformatics* 43, 1–11. [https://doi-](https://doi-org.stanford.idm.oclc.org/10.1002/0471250953.bi1110s43)
876 [org.stanford.idm.oclc.org/10.1002/0471250953.bi1110s43](https://doi-org.stanford.idm.oclc.org/10.1002/0471250953.bi1110s43)

877 Vietri Rudan, M., Barrington, C., Henderson, S., Ernst, C., Odom, D.T., Tanay, A., Hadjur, S.,
878 2015. Comparative Hi-C reveals that CTCF underlies evolution of chromosomal domain
879 architecture. *Cell Reports* 10, 1297–1309. <https://doi.org/10.1016/j.celrep.2015.02.004>

880 Vitillo, L., Anjum, F., Hewitt, Z., Stavish, D., Laing, O., Baker, D., Barbaric, I., Coffey, P.,
881 2023. The isochromosome 20q abnormality of pluripotent cells interrupts germ layer
882 differentiation. *Stem Cell Reports* 18, 782–797.
883 <https://doi.org/10.1016/j.stemcr.2023.01.007>

884 Weinreb, C., Raphael, B.J., 2016. Identification of hierarchical chromatin domains.
885 *Bioinformatics* 32, 1601–1609. <https://doi.org/10.1093/bioinformatics/btv485>

886 Wittkopp, P.J., Kalay, G., 2012. *Cis*-regulatory elements: Molecular mechanisms and
887 evolutionary processes underlying divergence. *Nature Reviews Genetics* 13, 59–69.
888 <https://doi.org/10.1038/nrg3095>

889 Wray, G.A., 2007. The evolutionary significance of *cis*-regulatory mutations. *Nature Reviews*
890 *Genetics* 8, 206–216. <https://doi.org/10.1038/nrg2063>

891 Yang, H., Ren, S., Yu, S., Pan, H., Li, T., Ge, S., Zhang, J., Xia, N., 2020. Methods favoring
892 homology-directed repair choice in response to CRISPR/Cas9 induced-double strand
893 breaks. *International Journal of Molecular Sciences* 21, 6461.
894 <https://doi.org/10.3390/ijms21186461>

895 Zhan, Y., Mariani, L., Barozzi, I., Schulz, E.G., Blüthgen, N., Stadler, M., Tiana, G., Giorgetti,
896 L., 2017. Reciprocal insulation analysis of Hi-C data shows that TADs represent a
897 functionally but not structurally privileged scale in the hierarchical folding of
898 chromosomes. *Genome Research* 27, 479–490. <https://doi.org/10.1101/gr.212803.116>

899 Zhang, J., Lee, D., Dhiman, V., Jiang, P., Xu, Jie, McGillivray, P., Yang, H., Liu, J., Meyerson,
900 W., Clarke, D., Gu, M., Li, S., Lou, S., Xu, Jinrui, Lochovsky, L., Ung, M., Ma, L., Yu,
901 S., Cao, Q., Harmanci, A., Yan, K.-K., Sethi, A., Gürsoy, G., Schoenberg, M.R.,
902 Rozowsky, J., Warrell, J., Emani, P., Yang, Y.T., Galeev, T., Kong, X., Liu, S., Li, X.,
903 Krishnan, J., Feng, Y., Rivera-Mulia, J.C., Adrian, J., Broach, J.R., Bolt, M., Moran, J.,
904 Fitzgerald, D., Dileep, V., Liu, T., Mei, S., Sasaki, T., Trevilla-Garcia, C., Wang, S.,
905 Wang, Y., Zang, C., Wang, D., Klein, R.J., Snyder, M., Gilbert, D.M., Yip, K., Cheng,
906 C., Yue, F., Liu, X.S., White, K.P., Gerstein, M., 2020. An integrative ENCODE resource
907 for cancer genomics. *Nature Communications* 11, 3696. [https://doi.org/10.1038/s41467-](https://doi.org/10.1038/s41467-020-14743-w)
908 [020-14743-w](https://doi.org/10.1038/s41467-020-14743-w)

- 909 Zhang, J., Sun-Waterhouse, D., Su, G., Zhao, M., 2019. New insight into umami receptor,
910 umami/umami-enhancing peptides, and their derivatives: A review. *Trends in Food*
911 *Science & Technology* 88, 429–438. <https://doi.org/10.1016/j.tifs.2019.04.008>
- 912 Zhang, Y., Liu, T., Meyer, C.A., Eeckhoutte, J., Johnson, D.S., Bernstein, B.E., Nusbaum, C.,
913 Myers, R.M., Brown, M., Li, W., Liu, X.S., 2008. Model-based analysis of ChIP-Seq
914 (MACS). *Genome Biology* 9, R137. <https://doi.org/10.1186/gb-2008-9-9-r137>
- 915 Zhu, J., Sanborn, J.Z., Diekhans, M., Lowe, C.B., Pringle, T.H., Haussler, D., 2007. Comparative
916 genomics search for losses of long-established genes on the human lineage. *PLoS*
917 *Computational Biology* 3, e247. <https://doi.org/10.1371/journal.pcbi.0030247>